National Aeronautics and
Space Administration

Synthetic transcriptomics for mouse spaceflight

Can generated RNA-seq improve FLT vs GC analysis within each tissue?

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Biological & Physical Sciences

QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

1,610 biological profiles

75 accessions

835 FLT

775 GC

ARCHS4 mouse

997,515 profiles audited

17,244 selected

20 tissues

- Mission, strain, material and assay differences can resemble a flight effect.
- The generator must preserve tissue and FLT/GC structure without memorizing samples.
- Statistical evidence must still come from observed OSDR profiles.

Sources: NASA OSDR Biological Data API; ARCHS4 mouse v2.5.

METHOD

We built a configurable bulk RNA-seq generation pipeline

Data scope, transforms, harmonization, training and conditioning can change without changing the evaluation contract.

Data + scope

OSDR API or ARCHS4
one or many studies
pooled or per tissue

Transform

raw / CPM / TPM
log + scaling
all genes / HVG / L1000

Harmonize

none or study z
ComBat / MBatch
MOBER adapter

Model + train

WGAN-GP or DDIM
OSDR or ARCHS4
pretrain + adapt

Condition

FLT/GC + tissue
study + material
available covariates

Selected branch

TPM / MaxAbs -> ARCHS4 DDIM -> OSDR adaptation. Conditions: tissue, FLT/GC, accession and material; no global batch correction.

A 463-row experiment plan and nine matched liver harmonization arms were screened before full training.

Generator sources: Vinas et al., Bioinformatics (2022); Lacan et al., BMC Bioinformatics (2026).

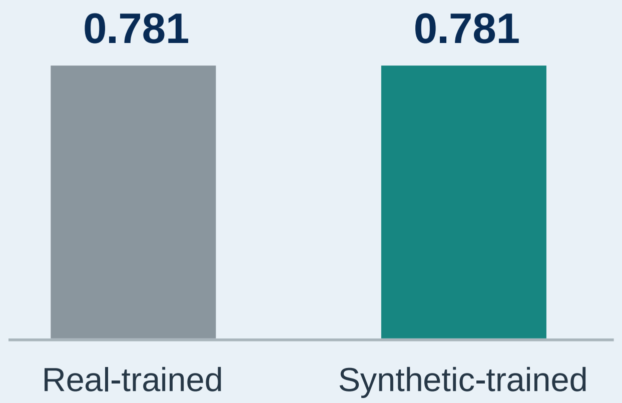
VALIDATION

DDIM matched expression and reduced separability

AA closer to 0.5 means a classifier has less success separating real and generated profiles; lower FD is better.

ARCHS4 tissue probe

Balanced accuracy on held-out real profiles



Metric	WGAN-GP	DDIM	Reading
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial accuracy	0.636	0.475	closer to 0.5
FD / real P95	0.144	0.074	lower

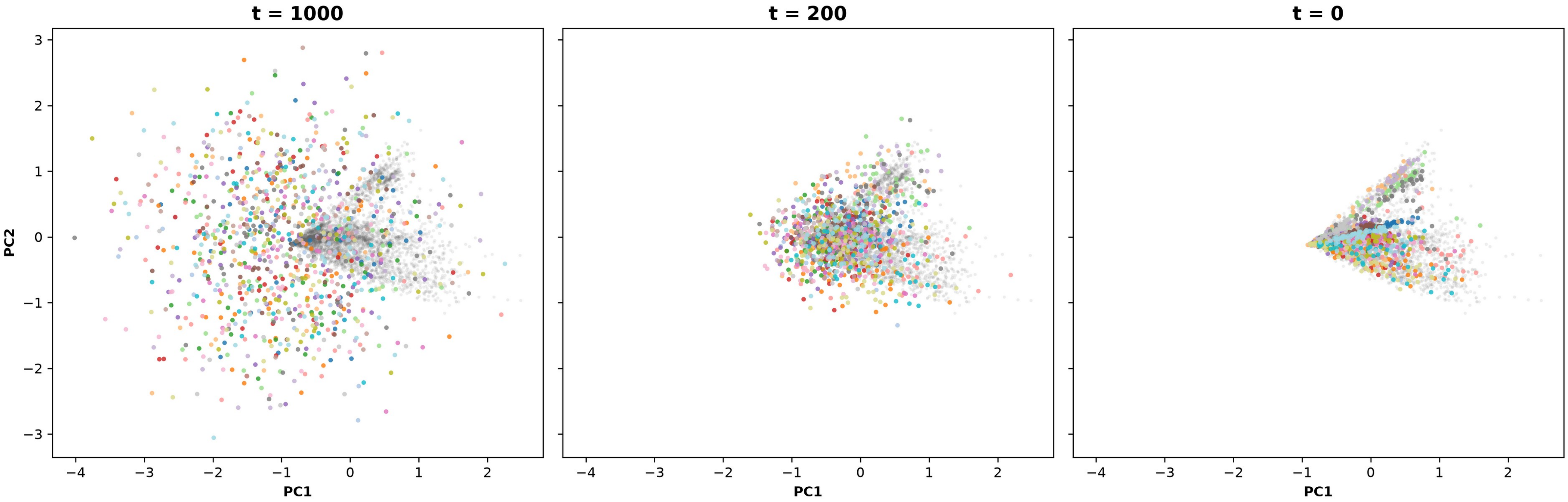
Use DDIM downstream

Lower separability and distributional distance, with comparable correlation and higher F1.

WGAN values use validation data and DDIM values use the stated OSDR test; they are not paired on one common split.

Diffusion learns tissue structure from noise

The same PCA axes follow 1,024 generated profiles through reverse diffusion.



t = 1000: noise

t = 200: partial structure

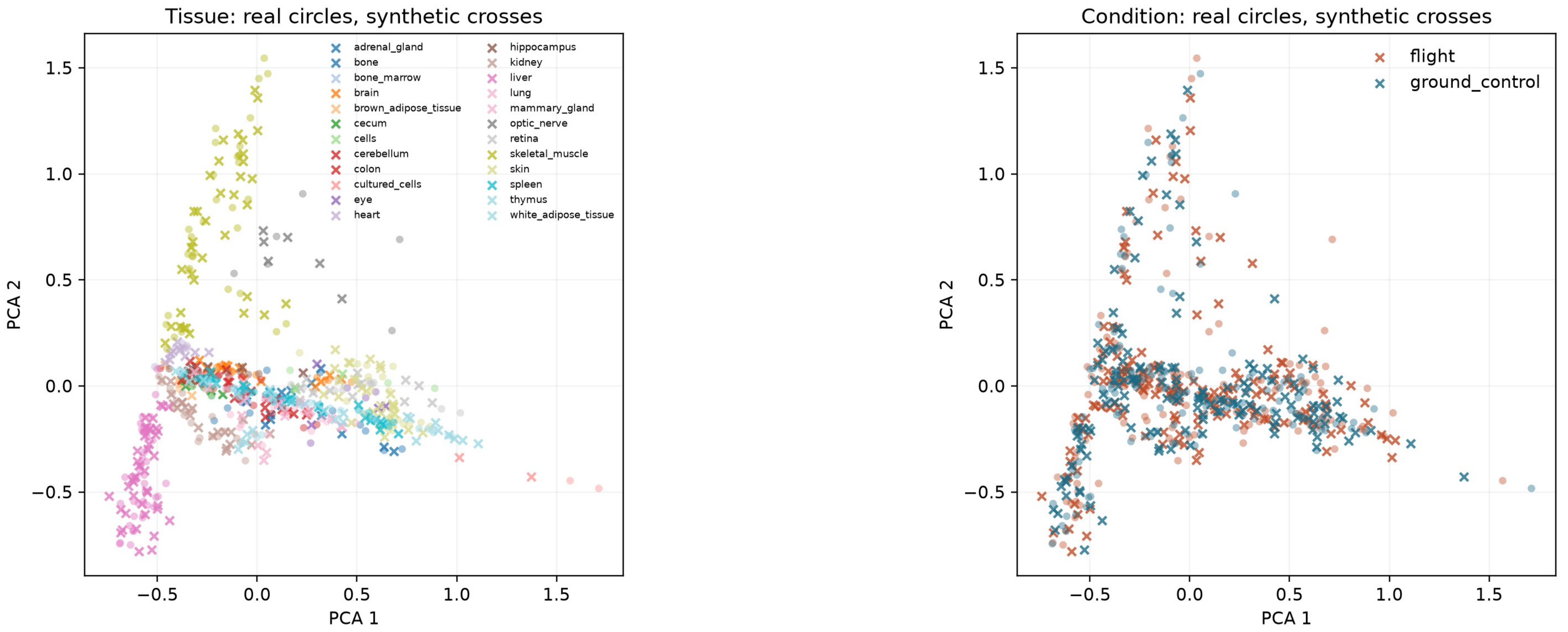
t = 0: tissue-conditioned profiles

Gray points are real ARCHS4 profiles; colors identify generated tissue conditions. PC1 and PC2 limits are shared across panels.

DIFFUSION OUTPUT

Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.



Interpretation: tissue structure dominates; FLT and GC are subtler. PCA is descriptive, so quantitative metrics determine validation.

PCA was fitted in the common locked OSDR expression space. This is PCA, not UMAP.

ANALYSIS

Synthetic profiles entered the analysis in five different ways

Every arm was judged on held-out real profiles before synthetic attribution was allowed.

Real OSDR
observed FLT / GC

DDIM generated
matched conditions

Real only
rank: real
fit: real

Generated only
rank: generated
fit: generated

Real + generated
rank: consensus
fit: equal weight

Guided, real fit
rank: consensus
fit: real

Guided, 5% synth.
rank: consensus
fit: real + 5%

Held-out real profiles

BA | AUROC | AP

select one eligible arm
for each tissue

Biology check FLT vs GC effects, random-effects meta-analysis and BH FDR use real OSDR profiles only.

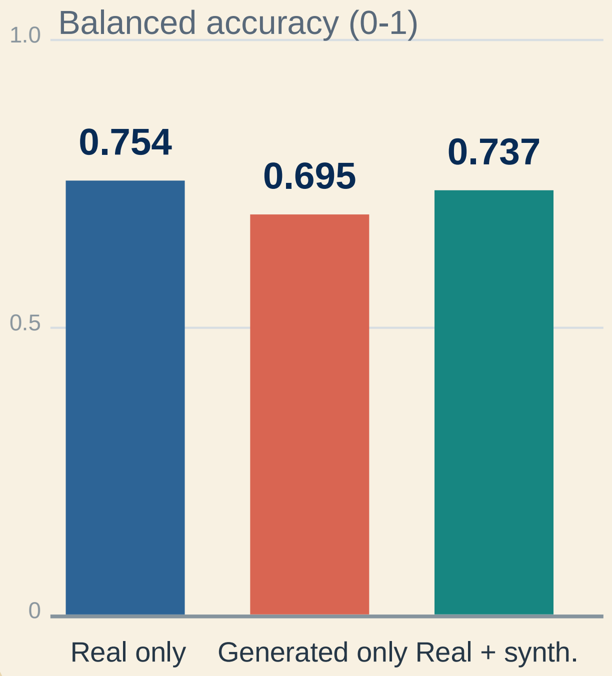
Synthetic profiles never increase animal n.

UTILITY

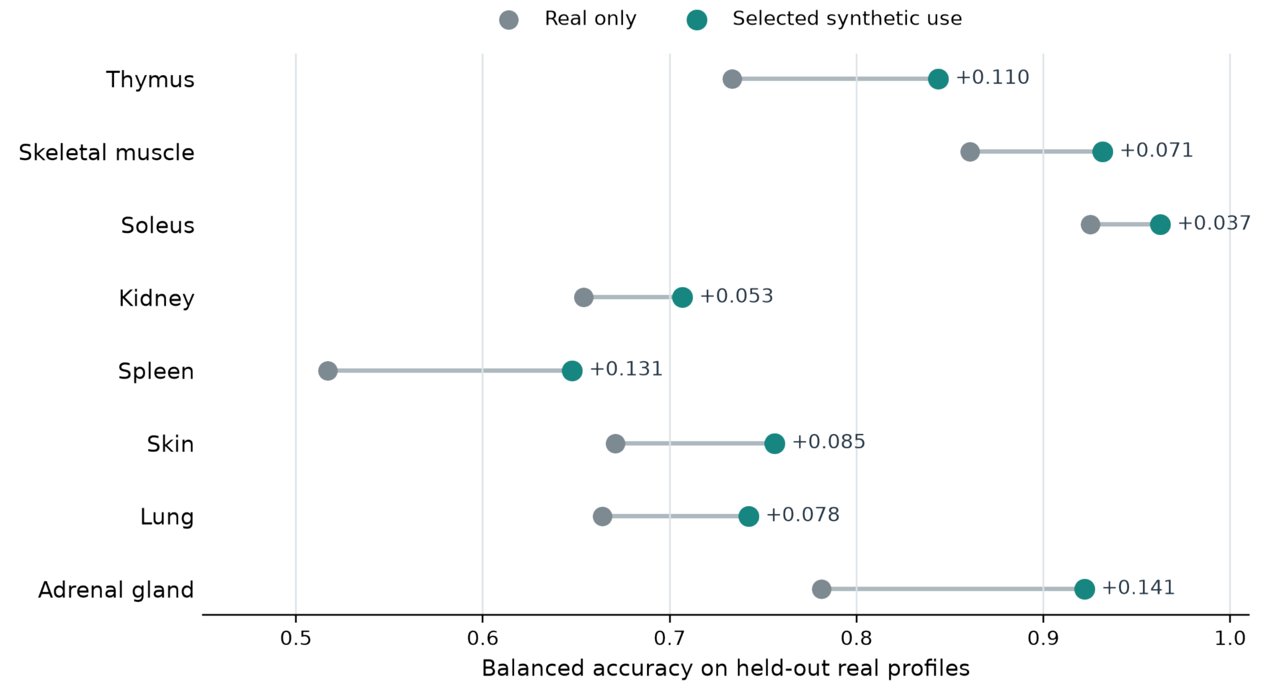
Pooling tissues hid useful signal

A single FLT/GC classifier did not improve, but tissue-specific synthetic use often did.

Pooled across tissues



Selected use within each tissue



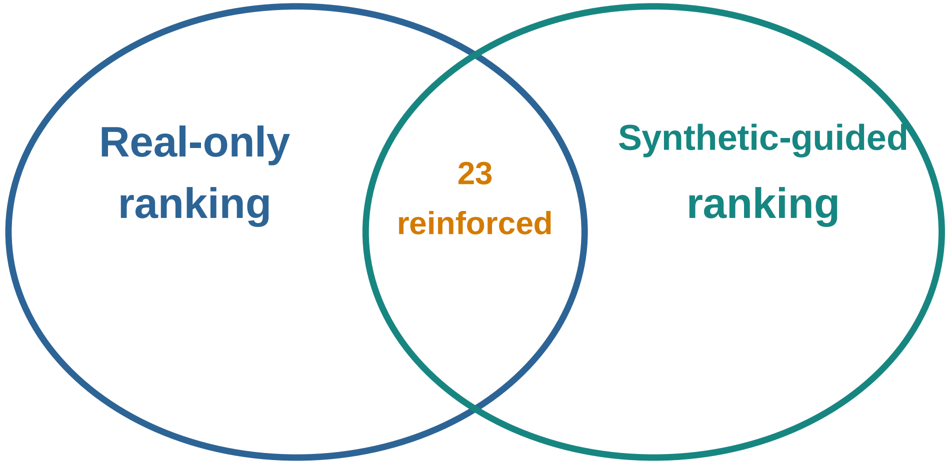
The useful arm differed by tissue.

Development results with held-out profiles from represented accessions; they do not measure transfer to a new mission.

Synthetic guidance changed ranking, not statistical evidence

Promoted and reinforced describe repeated feature selection; both require a supporting effect in real OSDR data.

Repeated stable selection



26 promoted
selected only with synthetic guidance

Real OSDR association test

- Estimate FLT vs GC inside each accession
- Combine accession effects with a random-effects model
- Apply BH FDR within each tissue

49 synthetic-informed associations

These are 49 tissue-gene associations among 459 BH-FDR rows. A gene can appear in more than one tissue.

LITERATURE INTERPRETATION

Selection and literature are separate dimensions

All 49 synthetic-informed associations received both a selection status and a literature interpretation.

1 Candidate
Gene + tissue +
FLT direction

2 Search
Spaceflight +
mechanism literature

3 Classify
Relation to
published evidence

4 Interpret
Recovery, extension,
or unmatched

Selection status

26 Promoted
Stable only with synthetic guidance

23 Reinforced
Stable with and without guidance

Literature interpretation

22	Aligning	Direct or same-tissue process agreement
19	Complementary	Related process or mechanism
4	Ambiguous	Mixed or context-dependent evidence
4	Unmatched	No sufficiently specific match found

A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.

Targeted review completed 2026-08-03; 5 direct transcript-level matches and a 33-source inventory are in Tables S16-S17.

ANALYSIS COVERAGE

All 27 completed tissue analyses were retained

The biological narrative narrows only after reporting synthetic-informed, real-only, and null outcomes.

10 Synthetic-informed

Promoted or reinforced BH-FDR gene

Adrenal Gland
Eye
Kidney
Skeletal muscle (pooled)
Skin
Spleen
Thymus
Gastrocnemius
Soleus
Tibialis Anterior

5 Real BH FDR only

Real association, no synthetic-informed selection

Heart
Liver
Retina
EDL
Quadriceps

12 No panel BH-FDR gene

No BH-FDR gene in the 974-gene panel

Bone
Bone Marrow
Brain
Brown Adipose Tissue
Cecum
Cerebellum
Colon
Hippocampus
Lung
Mammary Gland
Optic Nerve
White Adipose Tissue

Twenty-two canonical tissues plus EDL, gastrocnemius, quadriceps, soleus and tibialis anterior; complete counts in Table S12.

GENE INVENTORY

Synthetic-informed genes spanned 10 tissue analyses

All 49 associations passed BH FDR in real OSDR data; synthetic profiles affected prioritization, not the statistical test.

[P] promoted [R] reinforced — aligning — complementary — ambiguous — unmatched *Tag = selection | Gene color = literature*

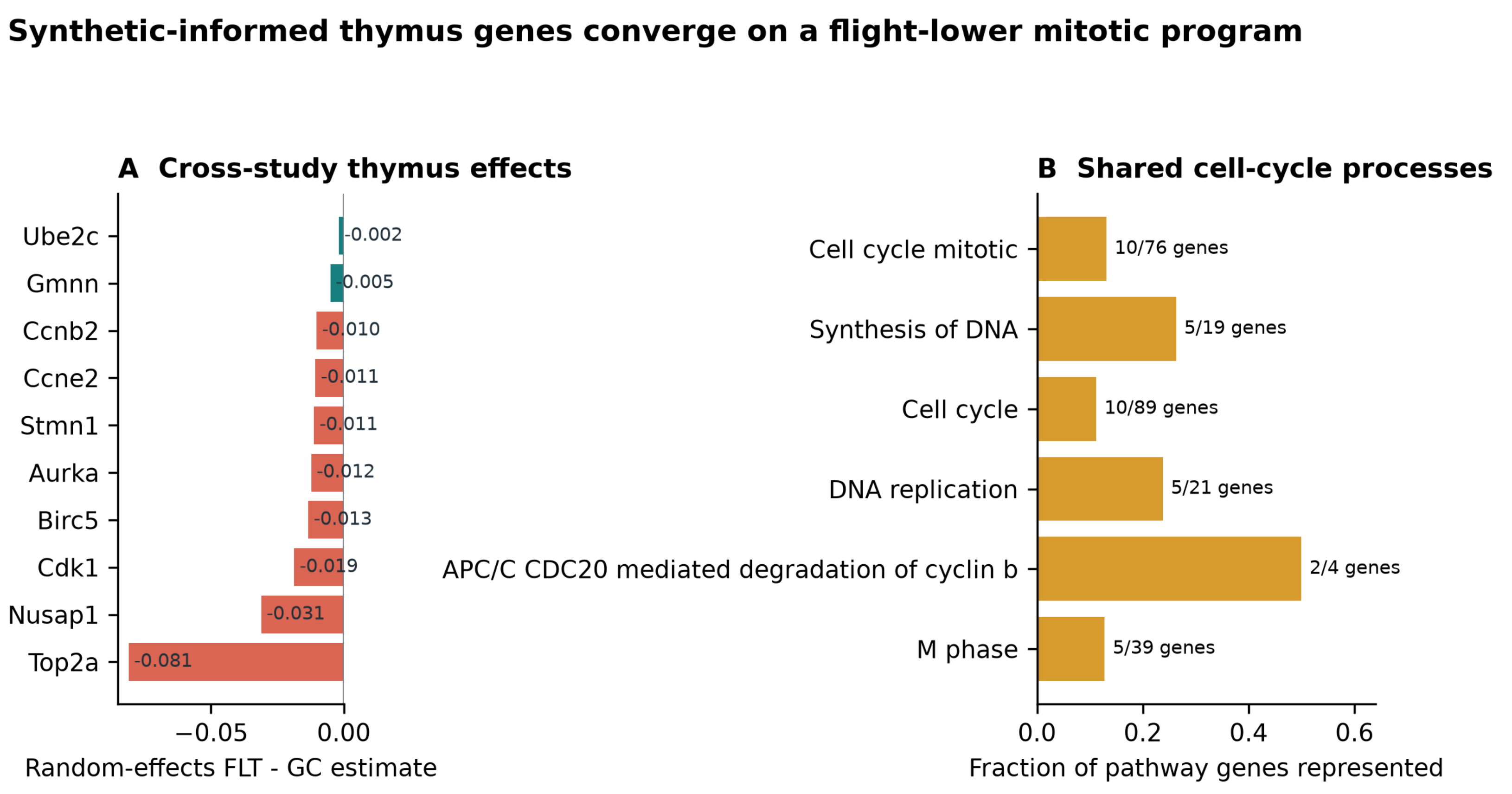
Thymus 16 genes	FLT higher	[P]*Hsd17b11*, [P]*Etv1*, [R]*Snx7*
	FLT lower	[P]*Nusap1*, [P]*Stmn1*, [P]*Birc5*, [P]*Cdk1*, [P]*Top2a*, [P]*Ccnb2*, [P]*Aurka*, [P]*Ccne2*, [P]*Kif20a*, [P]*Pcna*, [P]*Ccnf*, [R]*Ube2c*, [R]*Gmnn*
Spleen 4 genes	FLT higher	[P]*Rai14*, [P]*Myl9*, [P]*Ptprk*, [R]*Loxl1*
	FLT lower	*none*
Kidney 2 genes	FLT higher	[P]*Inpp4b*, [R]*Slc37a4*
	FLT lower	*none*
Gastrocnemius 2 genes	FLT higher	[P]*Nfkbia*
	FLT lower	[P]*Fhl2*
Eye 1 gene	FLT higher	*none*
	FLT lower	[R]*Klhl21*
Skeletal muscle (pooled) 12 genes	FLT higher	[R]*Sox4*, [R]*Cebpd*, [R]*Sh3bp5*, [R]*Prkcd*, [R]*Arid5b*, [R]*Sesn1*, [R]*Tle1*
	FLT lower	[P]*Klhl21*, [P]*Mapkapk5*, [P]*Reep5*, [P]*Itgb5*, [R]*Bphl*
Soleus 5 genes	FLT higher	[R]*Tpm1*
	FLT lower	[R]*Bdh1*, [R]*Ech1*, [R]*Bnip3*, [R]*Decr1*
Tibialis anterior 4 genes	FLT higher	[P]*Cebpd*, [R]*Cdkn1a*, [R]*St3gal5*, [R]*Bnip3*
	FLT lower	*none*
Adrenal gland 2 genes	FLT higher	*none*
	FLT lower	[P]*Psmb8*, [R]*Tspan4*
Skin 1 gene	FLT higher	[P]*Plscr1*
	FLT lower	*none*

[P]/[R] tags report repeated feature selection; gene color independently reports literature interpretation for all 49 associations.

THYMUS

Thymus points to lower proliferative renewal

The core panel recovers cell-cycle biology; two flight-higher genes extend the hypothesis.



16 synthetic-informed BH-FDR associations

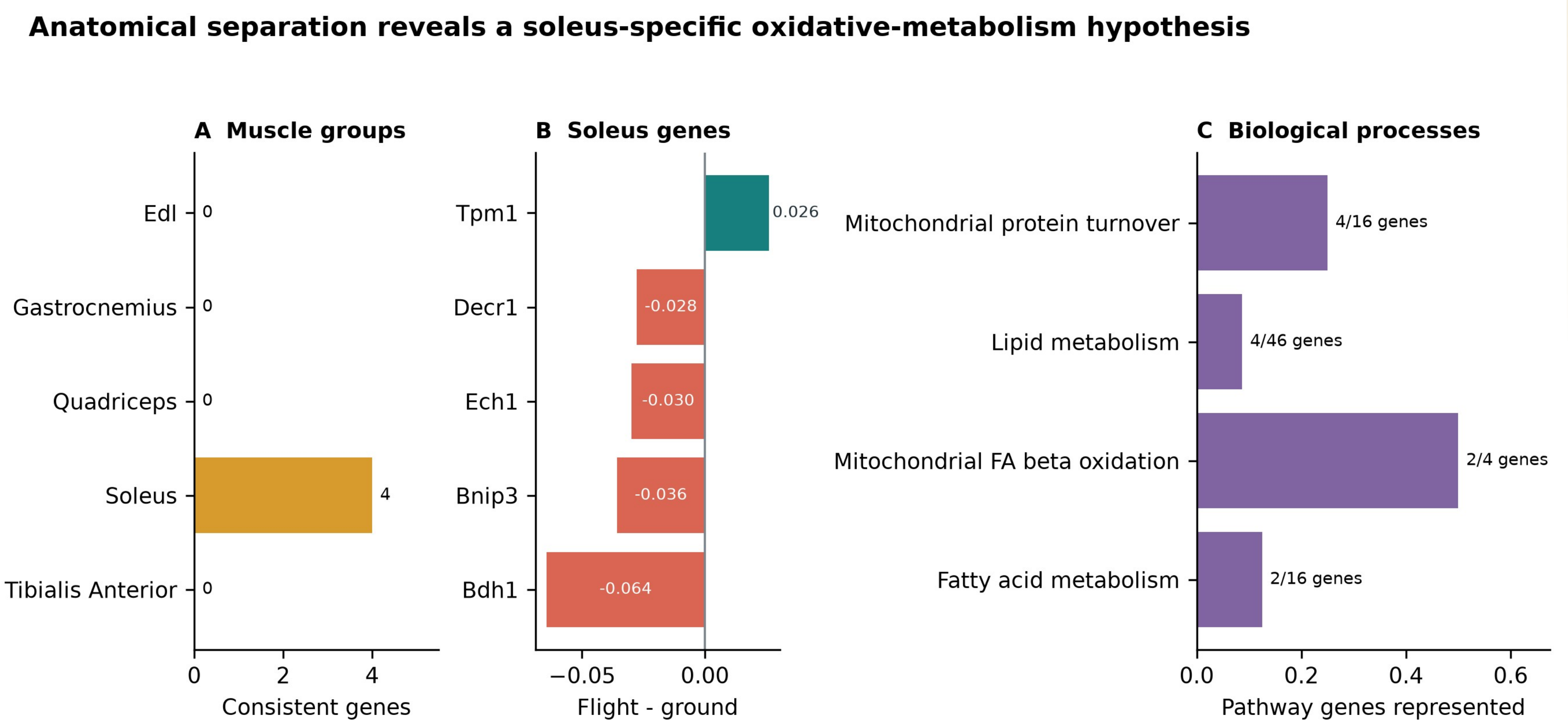
13 promoted | 3 reinforced

- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

Context: Gridley et al. (2013), Horie et al. (2019), Keenan et al. (2025), and Shi et al. (2026).

Soleus reinforces a mitochondrial and lipid program

This result strengthens an existing real-data pattern rather than promoting a new soleus gene.



0.925 -> 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

Literature context: Gambara et al. (2017) and Stein et al. (2002).

Each additional tissue defines a separate hypothesis

These results remain distinct; their placement on one slide does not imply shared biology.

Analysis unit	Synthetic-informed genes	Interpretation
Pooled muscle	12 genes: 4 promoted, 8 reinforced	Heterogeneous stress, differentiation, interferon and sialic-acid response; interpret with the anatomical muscle groups.
Kidney	[P] Inpp4b higher; [R] Slc37a4 higher	Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.
Spleen	[P] Rai14, Myl9, Ptprk higher; [R] Loxl1 higher	Adhesion, actomyosin and extracellular-matrix or immune-organization hypothesis; no coherent pathway enrichment.
Skin	[P] Plscr1 higher	Interferon-linked skin candidate within broader cell-cycle and DNA-repair responses; a single-gene result.

[P] promoted; [R] reinforced. All listed effects passed BH FDR in observed OSDR profiles.

ADDITIONAL FINDINGS II

Eye, adrenal and muscle-group results remain tissue-specific

Each row reports its own selected genes and the narrowest supported interpretation.

Analysis unit	Synthetic-informed genes	Interpretation
Eye	[R] Klhl21 lower	Process-level alignment with lower proliferation and cytokinesis in flight eye tissue; not an exact prior gene replication.
Adrenal gland	[P] Psmb8 lower; [R] Tspan4 lower	Literature-unmatched immune, proteostasis or tissue-composition candidates; no adrenal mechanism is established.
Gastrocnemius	[P] Nfkbia higher; [P] Fhl2 lower	NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.
Tibialis anterior	[P] Cebpd higher; [R] Cdkn1a, St3gal5, Bnip3 higher	Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.

[P] promoted; [R] reinforced. Interpretations follow the independent literature annotations in Table S16.

TAKEAWAYS

Synthetic data helped most as a tissue-specific prior

The generator created useful structure, not new biological replication.

1 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

2 Use

Pooled augmentation failed. Tissue-specific ranking and low-weight training were more useful.

3 Interpret

Thymus and soleus were clearest. Literature review separated recovery from complementary hypotheses.

Next test

Use independent samples and cell-resolved assays to test thymus proliferation, soleus metabolism and prioritized candidates from additional tissues.

Generated profiles never enter the biological sample count or the BH-FDR test.

Thank you

Questions?

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Code and manuscript	github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.